Probability	EPS (USD)
0.16	2.00
1.00	

1. What is the expected value of BankCorp's EPS for the current fiscal year?

   Solution:

   Following the definition of expected value, list each outcome, weight it by its probability, and sum the terms.

   $E(\text{EPS}) = 0.15(\text{USD}2.60) + 0.45(\text{USD } 2.45) + 0.24(\text{USD } 2.20) + 0.16(\text{USD } 2.00)$

   $= \text{USD}2.3405$

   The expected value of EPS is USD2.34.

2. Using the probability distribution of EPS from Exhibit 1, you want to measure the dispersion around your forecast. What are the variance and standard deviation of BankCorp's EPS for the current fiscal year?

   Solution:

   The order of calculation is always expected value, then variance, and then standard deviation. Expected value has already been calculated. Following the previous definition of variance, calculate the deviation of each outcome from the mean or expected value, square each deviation, weight (multiply) each squared deviation by its probability of occurrence, and then sum these terms.

   $$\begin{aligned}\sigma^2(\text{EPS}) &= P(2.60)[2.60 - E(\text{EPS})]^2 + P(2.45)[2.45 - E(\text{EPS})]^2 \\ &+ P(2.20)[2.20 - E(\text{EPS})]^2 + P(2.00)[2.00 - E(\text{EPS})]^2 \\ &= 0.15(2.60 - 2.34)^2 + 0.45(2.45 - 2.34)^2 \\ &+ 0.24(2.20 - 2.34)^2 + 0.16(2.00 - 2.34)^2 \\ &= 0.01014 + 0.005445 + 0.004704 + 0.018496 = 0.038785\end{aligned}$$

   Standard deviation is the positive square root of 0.038785:

   $\sigma(\text{EPS}) = 0.038785^{1/2} = 0.196939$, or approximately 0.20.

3 PROBABILITY TREES AND CONDITIONAL EXPECTATIONS

☐ formulate an investment problem as a probability tree and explain the use of conditional expectations in investment application

In investments, we make use of any relevant information available in making our forecasts. When we refine our expectations or forecasts, we are typically updating them based on new information or events; in these cases, we are using **conditional expected values**. The expected value of a random variable X given an event or scenario S is denoted $E(X \mid S)$. Suppose the random variable X can take on any one of n distinct